Program: **Diploma in Microelectronics**	
Course Code: **5498**	Course Title: **Simulation Lab 2**
Semester: **5**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To complement and reinforce the theoretical knowledge in CMOS VLSI Design.
- To implement CMOS circuits using PSPICE.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of VLSI design		VLSI Design	4
Implementation of combinational and sequential circuits	3044	Digital Electronics	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Develop simulation models to implement various combinational and sequential logic circuits.	12	Applying
CO2	Design VLSI Logic circuits with PMOS and NMOS logic families.	6	Applying
CO3	Design VLSI Logic circuits with CMOS logic families.	12	Applying
CO4	Design VLSI Logic circuits with Transmission gate and Pass Transistor logic.	9	Applying
	Lab Exam	6	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3			
CO2	3	3	3	3			
CO3	3	3	3	3			
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Develop simulation models to implement various combinational and sequential logic circuits.**		
M1.01	Simulate AND, OR, NAND, NOR, XOR, NOT gates using blocks	3	Applying
M1.02	Develop a model of Full adder and full subtractor	3	Applying
M1.03	Develop a model of 3-bit multiplexer and demultiplexer.	3	Applying
M1.04	Develop a model of D, T, and JK Flip-flop using NAND gates.	3	Applying
CO2	**Design VLSI Logic circuits with PMOS and NMOS logic families.**		
M2.01	Realize the following gates using NMOS logic NAND, NOR, OR, AND.	3	Applying
M2.02	Realize the following gates using PMOS logic NAND, NOR, OR, AND.	3	Applying
	Lab Exam	3	
CO3	**Design VLSI Logic circuits with CMOS logic families.**		
M3.01	Realize the following gates using CMOS logic NAND, NOR, OR, AND.	3	Applying
M3.02	Implement Boolean Expressions using CMOS logic.	9	Applying
CO4	**Design VLSI Logic circuits with Transmission gate and Pass Transistor logic.**		
M4.01	Realize the following gates using Pass Transistor logic NAND, NOR, OR, AND, NOT, XOR.	3	Applying

M4.02	Implement a 4*1 Mux using transmission gate logic.	3	Applying
M4.03	Implement a 8*1 Mux using transmission gate logic.	3	Applying
	Lab Exam	3	

****Suggested Open Ended Projects**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can-do open-ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation and a score of 15 marks. Students should prepare a separate report on open ended experiment of their choice.)

Example:

1. Implement 4:1 mux using CMOS logic.
2. Implement 3 input XOR using CMOS logic.

Text /Reference:

T/R	Book Title/Author
T1	Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits-Analysis & Design, McGraw-Hill, Third Ed., 2003
R2	Wayne Wolf ,Modern VLSI design, Third Edition, Pearson Education,2002.
R3	CMOS VLSI Design-A Circuits and Systems Perspective, Neil H.E Weste, David Harris, Ayan Banerjee, 3rd Edn, Pearson, 2009.
R4	Modern VLSI Design – Wayne Wolf, 3 Ed., 1997, Pearson Education.

Online Resources:

Sl.No	Website Link
1	https://onlinecourses.nptel.ac.in/noc24_ee29/preview
2	https://pages.hmc.edu/harris/cmosvlsi/4e/index.html
3	http://www.vlsi-expert.com/
4	https://semiwiki.com/